
MODEL SELECTION AND VALIDATION REPORT (remove_1_biomass model)

OBSERVATIONS:1732

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_1_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	6.82	0.36
remove_1_tric	0.15	0.03
remove_1_teve	-0.37	0.02
sbt_mean	0.27	0.08
sbt_sd	0.06	0.04
sst_sd	0.04	0.04
log_chla_mean	0.05	0.03
depth	0.05	0.04
log_sum_fishing	0.07	0.02

Dispersion parameter: 0.78

Spatiotemporal AR1 correlation (rho): 0.93

Matérn range: 483.58

Spatiotemporal marginal AR1 SD: 0.79

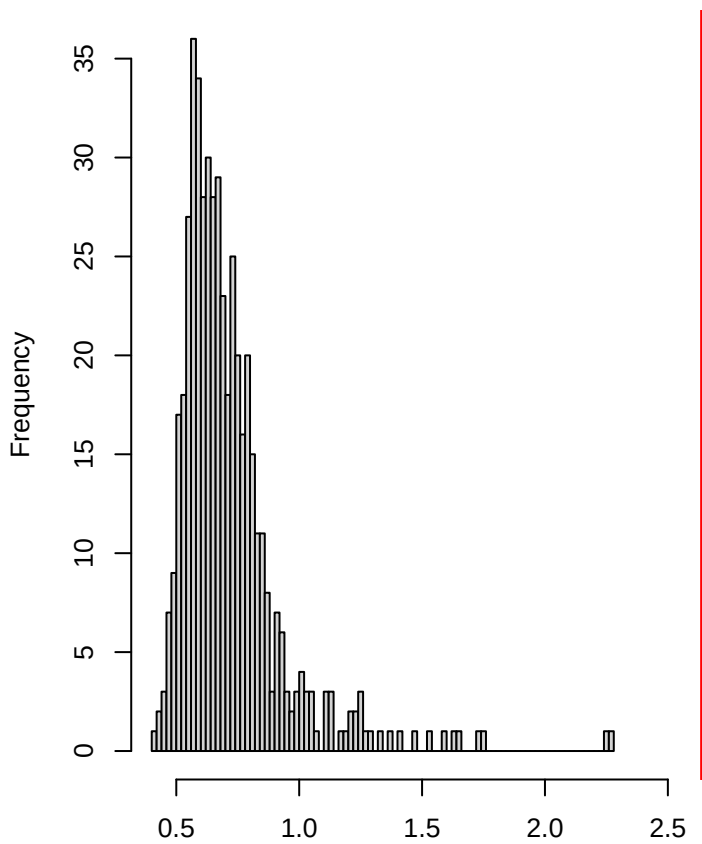
REML criterion at convergence: 13230.720

See ?tidy.sdmTMB to extract these values as a data frame.

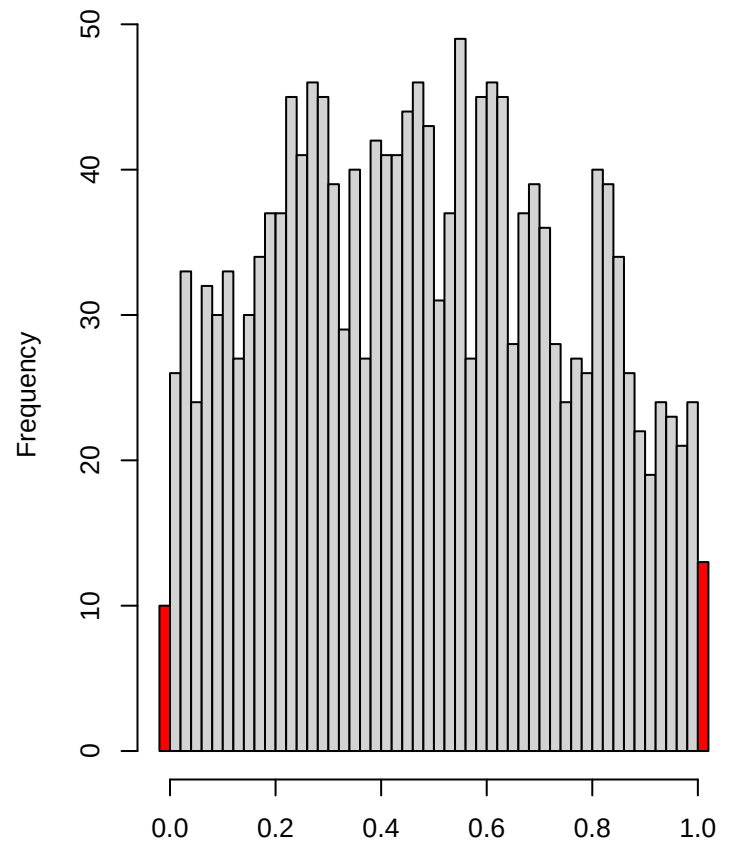
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	26487.44	2.898305
spatiotemporal_random_fields_ar1_with_anisotropy	15	26490.34	109.048982
spatiotemporal_random_fields_iid_with_anisotropy	14	26599.39	1.033940
spatiotemporal_random_fields_iid	12	26600.42	48.106986
spatial_random_fields	12	26648.53	3.617239
spatial_random_fields_with_anisotropy	14	26652.15	230.724172
error_only	10	26882.87	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



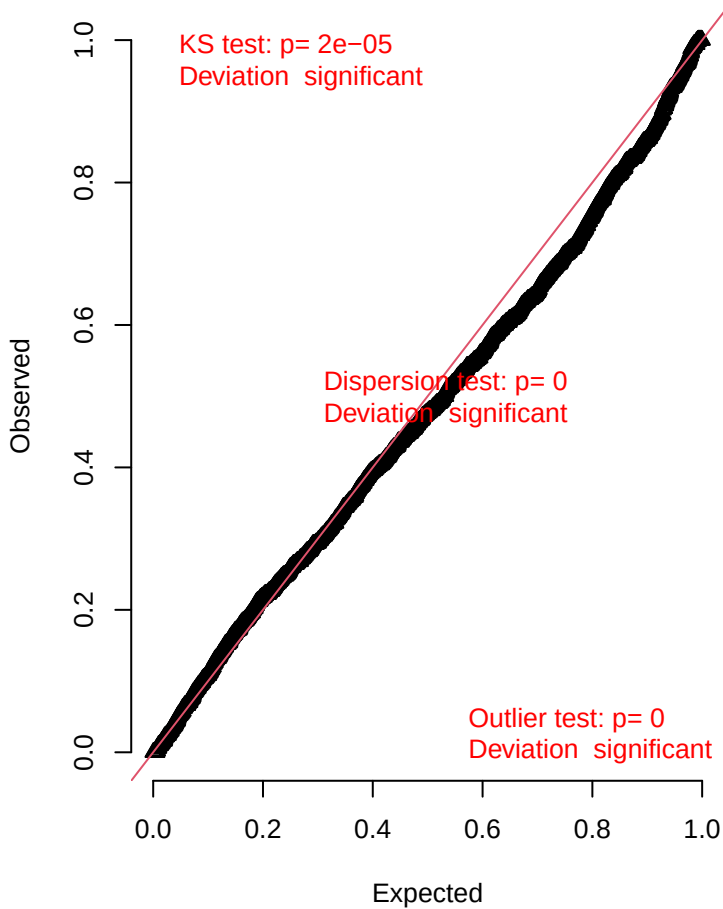
Outlier test significant



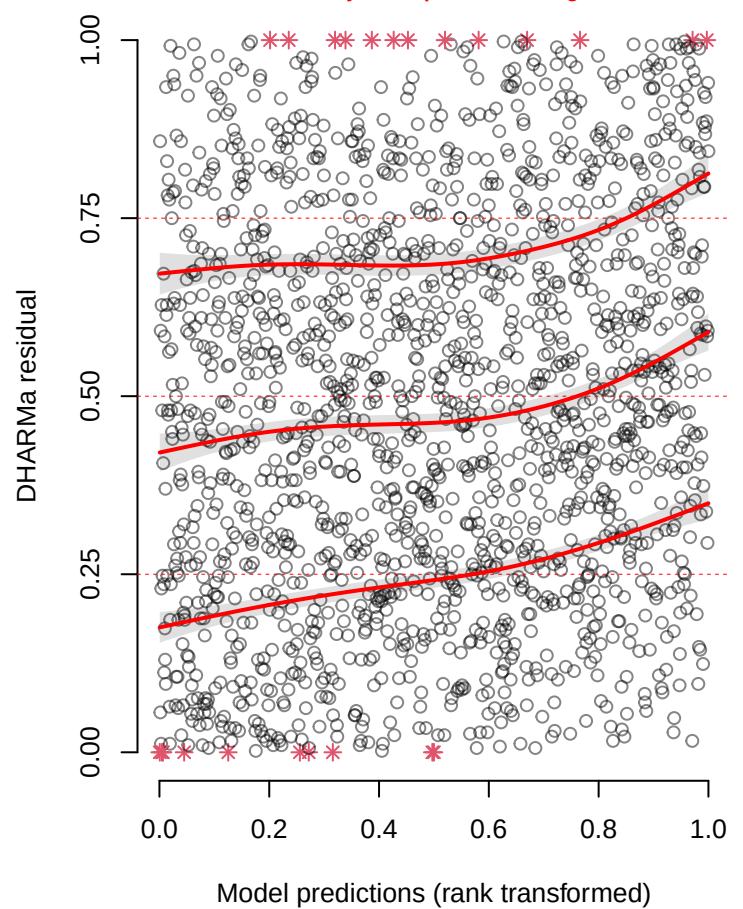
Simulated values, red line = fitted model. p-value (two.sided)

Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.038 | H1: Distance-based autocorrelation | p-value = 0

